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A policy brief on land use, the electrical grid, and the geography of inference.

By-Right Black*well*.

A manifesto for the Accessory Data Center Unit, and the application of California's ADU regulatory framework to artificial intelligence infrastructure.

> *Hyperscale concentrates the externality, the consent, and the rent in a handful of jurisdictions. Distribute all three.*

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EXECUTIVE SUMMARY

The hyperscale data center model concentrates the externalities of artificial intelligence infrastructure (power, water, land, transmission queue, particulate emissions from backup generation) in a small set of exurban jurisdictions whose residents had no meaningful role in the upstream decision. The workload composition that justified the model is changing. Training remains hyperscale-bound; inference, now the larger fraction of cumulative AI capital expenditure, does not. This brief proposes the *Accessory Data Center Unit* (ADCU): a sub-1,000 square foot, 60 to 100 kilowatt, ministerially permitted compute structure on residential parcels, governed by a regulatory framework modeled on California's 2019 ADU reforms and AB 1033 (2023). Four legislative actions are sufficient: by-right approval against an objective standard; ninety-day utility interconnection under FERC Order 2222 protocols; condominium-style separate conveyance; and Fannie Mae financing eligibility. The result is a distributed compute substrate that pays property tax, participates in wholesale demand response, recovers waste heat into domestic hot water, and routes the political economy of AI infrastructure through the only forum where land use politics has historically been adjudicable.

§ The hyperscale model has reached its political limit

For roughly a decade, the most consequential land use reform in the United States has not concerned skyscrapers, transit, or industrial siting. It has concerned the accessory dwelling unit. California's 2019 ADU framework, paired with SB 9 (2021) and AB 1033 (2023), produced more legal residential capacity in Los Angeles County than any state-administered affordable housing program over the same period. The mechanism was unsubtle. By-right approval of compliant units, on parcels homeowners already controlled, against an objective and pre-published standard, with separately conveyable title. The political economy of housing rerouted around the discretionary review process that NIMBY mobilization had previously weaponized. Supply expanded, prices softened, and the lot owner became a developer.

The same regulatory architecture, applied to a different building type, addresses the central political failure of the artificial intelligence infrastructure buildout.

The United States hyperscale fleet draws power on the order of forty gigawatts continuous; principal industry analysts project ninety gigawatts by 2030. Loudoun County, Virginia hosts approximately four gigawatts of operating load alone. Dominion Energy's 2024 Integrated Resource Plan requested authority to commission new gas-fired generation capacity to serve forecast data center demand, and the utility's interconnection queue is now measured in years.[1] Comparable patterns are documented in Maricopa County, Arizona; the Columbia River Gorge; and the Frankfurt-Amsterdam-London-Dublin axis. Three observations about this regime are no longer in serious dispute.

First, the externalities are spatially concentrated in jurisdictions that captured no meaningful share of the upstream consent process and frequently capture insufficient property tax revenue to compensate. Second, the political coalitions opposing further siting in those jurisdictions are now consolidated, well-funded, and effective; the Loudoun 2024 zoning amendments, the Mesa moratorium, and the Northern Virginia transmission litigation are leading indicators of a constraint that will tighten through the decade. Third, the workload composition that justified the original hyperscale topology is no longer the dominant workload composition.

Training requires tens of thousands of accelerators communicating at NVLink and InfiniBand bandwidths over a single low-latency fabric; the topology requirement is irreducible, and training will remain hyperscale-bound. Inference does not share the requirement. The decode phase of autoregressive generation on a dense transformer is memory-bandwidth-limited rather than compute-limited; arithmetic intensity per generated token is in the single-digit FLOPs-per-byte regime, the bottleneck is HBM throughput moving the key-value cache and weights to streaming multiprocessors, and the topology requirement collapses to a single accelerator package.[2] Continuous batching, paged attention, speculative decoding, and prefix caching preserve the property rather than disrupting it. Recommender system retrieval, the second largest workload by cumulative capital expenditure, is similarly local: sharded approximate-nearest-neighbor indices, replicated by region, with no all-reduce in the serving path. Inference is now the majority share of AI infrastructure spend in 2026 and is projected to exceed seventy percent of run-rate by 2028.[3]

> *“The fraction of compute that requires hyperscale colocation is declining, not increasing.”*

§ The ADCU specification

A reference Accessory Data Center Unit occupies between two hundred and eight hundred square feet of detached structure sited within the rear setback of a residential parcel. The unit houses two to eight current-generation accelerators in a single 4U or 8U chassis (the production successor to the Blackwell Ultra DGX form factor, projected by NVIDIA's published roadmap to be available in OEM-integrable configurations by 2027). Direct-to-chip liquid cooling carries thermal load to an exterior dry cooler mounted to the structure's leeward wall. Service is 100 kilovolt-amperes at 480 volt three-phase, drawn from a utility-coordinated upgrade of the parcel's existing service drop. Battery buffer capacity is fifty to one hundred kilowatt-hours, sized not for facility uptime but for participation in wholesale frequency regulation and demand response markets under FERC Order 2222 protocols.[4]

Target rack density is sixty to one hundred twenty kilowatts. Target power usage effectiveness is no greater than 1.08 at the meter; recovery of waste heat into a domestic hot water and radiant slab loop drives effective PUE below 1.0 from November through March in any climate zone north of Richmond, Virginia. Acoustic emission is capped at forty-five A-weighted decibels at the property line, achieved through standard isolator-mounted dry cooler installation and confirmed by post-installation field measurement. Compliance with NEC Article 645 (Information Technology Equipment) and ASHRAE TC 9.9 Class A2 environmental envelopes is required. Physical security relies on tamper-evident enclosures, attested boot, and confidential computing primitives at the silicon (Intel TDX, AMD SEV-SNP, ARM CCA); the threat model assumes an untrusted physical environment, which is the threat model under which the production chassis was already designed.

FIG. 1 · ADCU REFERENCE SPECIFICATION, V1

Accessory Data Center Unit / 60–100 kW class

FOOTPRINT	200 to 800 sq ft, detached, rear-yard sited
SERVICE	100 kVA · 480 V three-phase · utility-coordinated
COMPUTE	2 to 8 accelerators · single 4U / 8U chassis
COOLING	direct-to-chip liquid · exterior dry cooler
DENSITY	60 to 120 kW per rack
PUE	≤1.08 · effective <1.0 with DHW recovery, Nov–Mar, CZ4+
BUFFER	50 to 100 kWh storage · FERC 2222–eligible
ACOUSTIC	≤45 dB(A) at property line
CODE	NEC Article 645 · ASHRAE TC 9.9 Class A2
SECURITY	attested boot · TDX / SEV-SNP / CCA · tamper-evident enclosure

Networking is the most novel constraint, and the one most likely to be misread by readers accustomed to the hyperscale fabric. ADCUs do not federate at NVLink bandwidth; they federate at the bandwidth the workload tolerates. Continuous-batched inference on a single user session is embarrassingly parallel: one user, one node, no inter-node communication in the serving path. Recommender candidate retrieval traverses sharded and regionally replicated indices with no all-reduce. Small-model fine-tuning operates on a homogeneous local dataset. Federated and split learning regimes invert the traditional data-locality assumption: model parameters traverse the network, user data does not. Twenty milliseconds of last-mile residential latency, in this regime, is a feature; it places computation closer to the user than any centralized facility can achieve at any cost.

§ The ADU framework, applied

The 2019 California ADU framework, the 2021 SB 9 lot split provisions, and the 2023 AB 1033 separate-conveyance authorization together constitute the most successful land use reform of the past quarter century. Three structural features account for the success, and each translates without modification to the compute case.

By-right approval. The 2019 framework prohibits discretionary review of a compliant ADU. A unit meeting the objective standards (setback, height, parking, utility connection, fire access) is permitted as a matter of right, on a fixed timeline, by the local building official. The mechanism that NIMBY mobilization historically exploited (the public hearing, the design review, the conditional use permit) is removed from the application path entirely. A model ADCU ordinance applies the same logic. A unit meeting an objective enclosure standard (setback, height, maximum service amperage, maximum sound pressure level at the property line, conformance with NEC 645 and ASHRAE TC 9.9, attestation of confidential computing primitives) is ministerially permitted within thirty days. Concerns from neighboring property owners are resolved at the level of the standard, through the ordinary legislative process, not at the level of the individual application.

Prefabrication. The container-form-factor data center has been a commercial product for nearly twenty years. Sun Microsystems shipped Project Blackbox in 2006; Microsoft's IT Pre-Assembled Component (ITPAC) program productionized the form factor in the early 2010s; HPE Performance Optimized Datacenter, Vertiv SmartMod, and Schneider EcoStruxure currently sell turnkey edge units in the relevant capacity range.[5] The supply side exists. What does not exist is the demand-side legalization that would allow a homeowner, a condominium operator, or an inference provider acting on a homeowner's behalf, to procure a unit and site it on a residential parcel.

Separate conveyance. The pivotal feature of AB 1033 was the authorization to convey an ADU independently of the underlying parcel under a condominium-style instrument, which converted the ADU from a renovation expense into a financeable, securable, salable asset.[6] Adoption was opt-in at the municipal level and remains uneven; the legal scaffolding nevertheless exists. An ADCU under analogous treatment becomes a residential-zoned colocation asset class. The homeowner contributes the parcel and the service connection. An operator owns the chassis and the operating envelope. A hyperscaler or independent inference provider leases the capacity. The utility sees a dispatchable, geographically distributed, demand-response-eligible fifty kilowatt load that pays property tax and frequency regulation receipts. Each participant has a coherent reason to underwrite the transaction.

A residential-zoned colocation regime is functionally a securitization of latent grid capacity at the parcel level. The closest existing analogues are residential solar power purchase agreements and EV charging easements, both of which required enabling statutes before private capital underwrote at scale. The mortgage product is not novel; the asset class is not novel; the financing structure is not novel. The novelty is the legalization.

§ Steelmanning the principal objections

Five objections recur in informal canvassing of municipal planning officials, utility distribution engineers, and information security professionals. Each merits direct treatment.

Physical security. The objection asserts that valuable computational hardware in a residential setting is exposed to theft, tampering, or coercive physical access. The threat model is well precedented. Confidential computing primitives at the silicon (Intel TDX, AMD SEV-SNP, ARM CCA) provide attested execution under the assumption of a hostile physical environment. Tamper-evident enclosures and remote attestation at boot detect and respond to physical compromise. The hyperscale industry already operates under the equivalent threat model, because contemporary commercial colocation facilities are themselves multi-tenant environments with non-trivial insider risk.

Residential transformer overload. The objection is correct as a matter of distribution engineering. A typical residential pad-mounted or pole-mounted transformer is sized at twenty-five to fifty kilovolt-amperes; an ADCU at eighty kilowatts continuous would exceed it. The ordinance mandates utility coordination and any necessary service upgrade as a precondition of permitting, with upgrade cost borne by the applicant. In exchange the utility acquires a dispatchable, individually metered load at the edge of the distribution feeder, which utility distribution planners have been requesting from regulators for fifteen years. The transformer upgrade is a feature of the regime, not a defect.

Acoustic emission. The dry cooler is the dominant noise source. Standard isolator-mounted installation, conservative blade-tip speed selection, and shielding bring sound pressure levels at the property line below forty-five A-weighted decibels, which is at parity with a residential heat pump condenser of comparable capacity. The ordinance specifies the limit, the manufacturer specifies the equipment, the building official verifies post-installation field measurement. None of these steps is novel.

Stranded asset risk. Accelerator generations turn over on twenty-four to thirty-six month cycles; a fifteen-year mortgage will outlast four to six chassis refreshes. The applicable accounting treatment separates the durable asset (the slab, the service drop, the cooling loop, the network handoff, the enclosure) from the consumable asset (the chassis). The ADCU is a long-lived shell hosting short-lived silicon, on the same depreciation logic that distinguishes a Manhattan office tower from its tenants' workstations. Operator agreements assume the refresh schedule and price the lease accordingly.

Distributional concern. The most common political objection is that ADCUs constitute a transfer to the homeowner class at the expense of renters, multifamily residents, and exurban communities currently bearing the externalities of the hyperscale regime. The objection inverts the actual distribution. The hyperscale-only regime concentrates the externality (in exurban jurisdictions), the consent (in a handful of local planning commissions), and the rent (in a small number of vertically integrated operators). The ADCU regime distributes all three. Multifamily and commercial parcel applications follow under the same statutory framework with proportionate adjustments to maximum capacity per parcel.

§ The legislative ask

Three pages of statute, accompanied by an enclosure standard schedule co-developed by ASHRAE TC 9.9 and the American Institute of Architects. A FERC Order 2222 implementation rule extending interconnection timelines to ninety days for compliant ADCUs. A Fannie Mae product specification permitting first-lien financing of the durable enclosure and operating lease treatment of the consumable chassis. A federal income tax provision that treats the household as a productive infrastructure investor. The model ordinance follows.

MODEL ORDINANCE · THREE CLAUSES

An Act to Authorize Accessory Data Center Units

§ 01 ***By-Right Approval.*** No municipality shall require discretionary review for any Accessory Data Center Unit meeting the objective standards set forth in Schedule A, including setback, height, maximum service amperage, maximum sound pressure level at the property line, and conformance with NEC Article 645 and ASHRAE TC 9.9. Compliant units shall be ministerially permitted within thirty days of a complete application.

§ 02 ***Interconnection & Grid Services.*** The serving utility shall complete interconnection studies and effectuate any necessary service upgrade for a compliant Accessory Data Center Unit within ninety days of application, modeled on FERC Order 2222 protocols. The unit shall be eligible to participate in wholesale demand response, frequency regulation, and capacity markets on equal terms with utility-scale resources.

§ 03 ***Separate Conveyance & Finance.*** An Accessory Data Center Unit may be conveyed independently of the underlying parcel, recorded under a condominium-style instrument modeled on California AB 1033, and shall be eligible for federally-backed mortgage financing on equal terms with primary residential structures.

The twentieth century underwrote a settlement with the American single-family parcel: in exchange for the front yard, the driveway, and the detached garage, productive economic activity occurred elsewhere, ideally out of sight. That settlement is being unilaterally reopened by the physical demands of an information infrastructure that has outgrown its preferred siting envelope. The terms are negotiable at the lot line, with a meter, a check, and a thermostat preheated by a language model; or they are negotiable in Loudoun, Maricopa, and the Columbia River Gorge, on terms set by the parties presently bearing the cost. Only the first option scales.

> *“By-right Blackwell. In every backyard.”*

Notes

1 Loudoun County's "Data Center Alley" hosts the largest single concentration of operating data center capacity globally, with operating load on the order of four to five gigawatts and announced expansion of comparable scale. Dominion Energy's 2023 and 2024 Integrated Resource Plans document the transmission queue and request authority for additional gas-fired generation to serve forecast demand. See PJM Interconnection's quarterly load forecasts and Virginia State Corporation Commission Case PUR-2024-00099.

2 Arithmetic intensity of decode-phase autoregressive inference on a dense transformer falls in the single-digit FLOPs-per-byte regime, well below the ridge point of any current accelerator's roofline. The binding constraint is HBM throughput moving the KV cache and weights to streaming multiprocessors. Prefill is compute-bound, decode is bandwidth-bound. The asymmetry is the architectural reason inference does not require a hyperscale fabric. See Williams, Waterman & Patterson (2009) on roofline modeling and Pope et al. (2022) on transformer inference scaling.

3 Inference share of total AI infrastructure capital expenditure is documented in NVIDIA, AMD, and Broadcom 10-K disclosures across fiscal years 2024 and 2025, as well as McKinsey Global Institute's *Generative AI Capital Stack* series. The seventy-percent run-rate projection reflects the consensus of major equity research desks as of Q1 2026.

4 FERC Order 2222 (2020) requires regional transmission organizations to permit aggregations of distributed energy resources to participate in wholesale electricity markets. ISO compliance has been slow and uneven; an ADCU regime would slot into the existing framework rather than requiring a parallel one. See FERC Docket No. RM18-9-000 and ISO compliance filings 2021 to 2025.

5 Container-form-factor data center products have been commercially available since Sun Microsystems' Project Blackbox (2006); Microsoft's ITPAC program productionized the form factor in the early 2010s; HPE Performance Optimized Datacenter, Vertiv SmartMod, and Schneider EcoStruxure occupy the relevant capacity range. The hardware lineage substantially predates the policy gap it now exposes.

6 California AB 1033 (2023) authorized municipalities to permit the separate conveyance of ADUs as condominiums, completing the legal trilogy with SB 9 (2021) lot splits and the 2019 by-right ADU framework. Adoption is opt-in at the municipal level; uptake has been uneven, but the enabling statute is in force and the financing infrastructure is being built. See California Government Code §§ 65852.2 and 65852.26.

COLOPHON

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